

Dora Elalaoui-Pinedo

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My research interests center on surface processes across Mars, Earth, and Europa — including geomorphology, mineralogy, and the evolution of ice sheets and glaciers. I am interested in using imaging spectroscopy, remote sensing, field geology, and change detection techniques, along with data science and machine learning methods, to study active planetary surfaces and assess planetary habitability. I am also drawn to planetary science mission instrumentation.

EDUCATION

University of Arizona

Department of Geosciences, Lunar & Planetary Laboratory, Department of Mathematics

Tucson, AZ
Expected May 2027

- B.S. Planetary Geosciences, B.S. Statistics and Data Science
- Minors in Astrobiology, Leadership Studies & Practice

Relevant Courses: Mineralogy, Petrology, Structural Geology, Mars, Intro to Statistical Computing

Cumulative GPA: 3.70

RESEARCH & WORK EXPERIENCE

University of Arizona, Undergraduate Student Researcher

Tucson, AZ
Aug. 2026–Present

- Produce digital terrain models of the South Polar Layered Deposits (SPLD) of Mars to catalog exposed strata, under Dr. Shane Byrne.
- Support the efforts to constrain the recent climate expressed in the geomorphic and geological features of the SPLD.

University of Arizona, HiRISE Student Research Technician

Tucson, AZ
May 2024–Present

- Create digital terrain models (DTMs) of the surface of Mars with stereophotogrammetry.
- Collaborate with researchers to explore geological processes on Mars using models.
- Teach and support attendees in a biannual planetary photogrammetry workshop.

SETI Institute, REU Intern

Characterizing Morphological and Mineralogical Variability of Interdunes at Olympia Undae, Mars

Mountain View, CA
Jun. 2026–Aug. 2026

- Analyzed morphological and mineralogical patterns in Martian interdunes using CRISM and HiRISE data, under Dr. Janice Bishop and Dr. Lori Fenton.
- Mapped interdune morphology using JMARS and collected mineral spectra with ENVI.
- Identified spectral evidence of iron sulfates, including potential rozenite, in interdunes.
- Investigated spectral evidence of evaporite minerals in interdunes to uncover aqueous processes shaping Mars's north polar climate history.

University of Arizona, NASA Space Grant Undergraduate Intern and Mentor

Mapping Enigmatic Pits in the North Polar Layered Deposits (NPLD) of Mars

Tucson, AZ
Aug. 2023–May 2025

- Investigated meter-scale pits using HiRISE imagery under Dr. Sarah Sutton.
- Analyzed more than 400 HiRISE images and located and mapped over 2,000 pits using QGIS.
- Created DTMs to conduct a temporal analysis of potential pit morphology changes.
- Identified evidence of active surface processes, potential faults, and ice sublimation or deposition in the NPLD to help uncover Mars's climate history.
- Mentored first-year Space Grant Interns through the research process and created networking experiences.

PUBLICATIONS

1. **Elalaoui-Pinedo, D.E.**, Sutton, S.S., et al. (in preparation). Enigmatic Pits in the North Polar Layered Deposits of Mars: Mapping and Seasonal Analysis.
2. Kelly, E., Andrews, S., [...] **Elalaoui-Pinedo, D.E.**, et al. (submitted). HCN and HNC 1–0 Emission in the Central Taurus Molecular Cloud. *Publications of the Astronomical Society of the Pacific*.

CONFERENCE PRESENTATIONS

(* = Oral)

1. ***Elalaoui-Pinedo, D.E.**, Bishop, J.L., Fenton, L.K., et al. (2026). Analyzing Patterns in Morphology and Mineralogy of Interdunes at Olympia Undae, Mars. *Geological Society of America Connects 2026*, Abstract #20070.

2. **Elalaoui-Pinedo, D.E.**, Sutton, S.S., Okubo, C., Byrne, S. (2025). Investigating Enigmatic Pits in the North Polar Layered Deposits of Mars. *Lunar and Planetary Science Conference LVI*, Abstract #2486.
3. ***Elalaoui-Pinedo, D.E.** & Sutton, S.S. (2025). Investigating Enigmatic Pits in the North Polar Layered Deposits of Mars. *Arizona NASA Space Grant Statewide Student Research Symposium XXXIV*.
4. ***Elalaoui-Pinedo, D.E.** & Sutton, S.S. (2024). Mapping Enigmatic Pits in the North Polar Layered Deposits of Mars. *Arizona NASA Space Grant Statewide Student Research Symposium XXXIII*.

TECHNICAL SKILLS & LANGUAGES

- Python, R, MATLAB, QGIS, SOCET SET, ENVI/IDL, JMARS, LaTeX, Microsoft Office (Word, Excel, PowerPoint, etc.), remote sensing, spectral analysis, image analysis, data analysis, scientific writing.
- Native in English and Spanish.

LEADERSHIP & TEACHING EXPERIENCE

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS), President, University of Arizona

Tucson, AZ
Apr. 2024–Present

- Organize science education outreach at underserved Tucson public schools, leading 10 visits since 2024.
- Direct monthly community-building events and professional development programming.
- Guide chapter meetings and connect with other campus organizations to grow member engagement and participation.
- Lead and participate in environmental conservation volunteer efforts alongside local advocacy groups.

Housing and Residential Life

Resident Assistant, University of Arizona

Tucson, AZ
Jul. 2023–May 2026

- Served as a mentor for ~30 residents, fostering community through regular one-on-one check-ins and dorm-wide events on topics such as academic success and diversity.
- Connected residents to campus resources to help them navigate challenges during their time as a student.
- Responded to and de-escalated resident conflicts, crises, and policy violations to ensure resident safety and wellbeing.

Blue Chip Leadership

Peer Mentor, University of Arizona

Tucson, AZ
Aug. 2024–May 2025

- Facilitated a leadership development curriculum for a cohort of 20 first-year students, covering topics such as goal setting, leadership styles, and community engagement.
- Mentored first-year students through academic and personal challenges, connecting them to campus resources and helping them build self-confidence.
- Led weekly group sessions and community service projects to apply lessons on identifying systemic obstacles to opportunity and translating them into actionable, student-led initiatives.

Invited Guest Speaker

Topics in Planetary Science (PTYS 395B), University of Arizona

Sep. 2026

- Delivered final SETI Institute REU presentation, “Characterizing Morphological and Mineralogical Variability of Interdunes at Olympia Undae, Mars.”
- Facilitated a discussion and Q&A session with students regarding the REU application process, timeline, and experience.

HONORS & AWARDS

Facilitated Graduate Admissions Program (F-GAP Scholar), Math Alliance at the American Mathematical Society	2026-2027
William Y. Vélez Scholarship	2026-2027
Hispanic Alumni Tuition Scholarship	2022-2026
Joel Valdez Scholarship	2022-2026

PROFESSIONAL ORGANIZATIONS & GROUPS

Society for Advancement of Chicanos/Hispanics & Native Americans in Science (SACNAS)	2022
Epsilon Eta Pre-Professional Environmental Fraternity, Tau Chapter	2025
Geological Society of America	2026
GeoLatinas	2026